Assignment 1: Results Matrix

BITS F464 Machine Learning — Flat vs. Hierarchical Classification From Scratch

Anuran Calls (MFCCs); 80/20 split stratified on Species (seed 42): 5,756 train / 1,439 test rows. All accuracies in %.

Table 2: Results matrix (flat models predict Species)

Method	Criterion	Overall accuracy (Species)		Family acc. †
		Train	Test	Test
1. Single Decision Tree	Gini	100.00	94.93	96.25
	Entropy	100.00	95.41	96.80
2a. Bagging	Gini	99.97	96.66	97.50
	Entropy	99.98	96.87	97.50
2b. AdaBoost	Gini	97.52	95.34	96.73
	Entropy	96.35	94.16	95.00
3. Random Forest	Gini	100.00	97.78	98.40
	Entropy	100.00	97.64	98.19
4. Hierarchical Decision Tree	Gini	100.00	93.82	Router: 95.07
	Entropy	100.00	94.65	Router: 95.62
5. Gaussian Naive Bayes	—	91.47	90.34	91.94

† For flat models, the derived family accuracy (each predicted species mapped to its family). For the Hierarchical rows, the router accuracy; the Overall accuracy columns hold the final species accuracy.

Table 3: Supplementary hierarchical failure analysis (test set, %)

Criterion	Router acc.	Oracle acc.	(A) Router right, expert wrong	(B) Router wrong
Gini	95.07	98.26	1.25	4.93
Entropy	95.62	98.33	0.97	4.38

Check: overall accuracy = router accuracy – (A), i.e. Gini 95.07 – 1.25 and Entropy 95.62 – 0.97, matching the hierarchical test accuracies above.